CS59300 Final Project Proposal

Gustavo Pereira Corsi da Silva
Purdue University Fort Wayne
pereg03@pfw.edu

Abstract

Hate speech detection systems are commonly used on social media platforms to automatically flag harmful content. However, research has shown that these systems may unfairly label language associated with certain communities or dialects as toxic, even when it is not intended to be offensive. This project will examine whether hate speech classifiers produce higher false positive rates for identity-related terms or dialectal language. I plan to train supervised text classification models on standard hate speech datasets and evaluate their performance using accuracy and F1-score. In addition to overall performance, I will analyze subgroup metrics such as false positive rate differences across identity groups. Through subgroup analysis and counterfactual testing, this project aims to better understand how model design and dataset composition may contribute to biased outcomes in automated content moderation systems.

1 Introduction

The motivation for this project is to investigate fairness issues in hate speech classifiers and analyze how language variety, dialect, and community-specific terms influence model predictions. By examining bias patterns, this project aims to better understand the ethical tradeoffs between content moderation and fairness.

Primary Ethical Question: Do hate speech detection models unfairly classify language associated with specific communities or dialects as toxic or hateful?

2 Problem Statement

Given a hate speech classification model trained on standard datasets, evaluate whether prediction errors (false positives and false negatives) are disproportionately distributed across language varieties, identity-related terms, or dialects. Specifically, this project will investigate whether sentences containing identity terms are more likely to be classified as toxic, whether dialectal language, is more likely to be misclassified, and whether the model makes more mistakes for certain demographic language groups.

3 Proposed Method

I plan to begin with supervised text classification approaches for hate speech detection. Baseline models will include logistic regression with TF-IDF features, support vector machines, and potentially Naive Bayes for comparison. Advanced models will consist of fine-tuned transformer-based models, such as BERT or DistilBERT, and pre-trained toxicity detection APIs may also be used for comparison. The objective is not only to achieve strong predictive performance but also to examine how different model architectures influence fairness outcomes.

In addition to standard model training, fairness-aware strategies will be explored. These may include adjusting decision thresholds across subgroups, reweighting or debiasing the training data, and augmenting data to better represent underrepresented dialects. Together, these approaches will allow me to compare performance across models while evaluating potential bias mitigation strategies.

4 Datasets

The project will use established hate speech and toxicity detection datasets, including the Hate Speech and Offensive Language Dataset (Davidson et al., 2017), the Jigsaw Toxic Comment Classification Dataset (Jigsaw, 2018), the Civil Comments Dataset (Borkan et al., 2019), and the Hatebase Dataset (Hatebase, 2012).

The Civil Comments dataset is particularly valuable for this study because it includes identity annotations (e.g., references to gender, religion, race, and sexual orientation), which enable subgroup-based fairness evaluation and the measurement of performance disparities across demographic groups. These identity labels support the computation of subgroup-specific metrics such as false positive rates, false negative rates, and F1 scores.

If available, supplementary datasets containing dialectal language, such as corpora associated with African American English (AAVE), may be incorporated to facilitate dialect-based bias analysis. Incorporating such datasets would allow for comparison of model behavior across linguistic varieties and help assess whether classifiers disproportionately flag dialectal language as toxic. Additional datasets may be included to enable deeper subgroup evaluation and more robust bias analysis.

links to databases.

5 Evaluation Plan

Model performance will be assessed using standard classification metrics including accuracy, precision, recall, and F1-score. Subgroup-based evaluation metrics will also be computed, such as false positive rate and false negative rate for each subgroup, equality of opportunity gaps, and disparities in F1-score across identity groups. A successful project will train a competitive hate speech classifier, identify measurable fairness disparities across subgroups, quantify the magnitude of these disparities, and compare bias patterns across different model architectures.

6 Planned Analysis

Throughout the project, additional analyses will be conducted to understand sources of bias. For example, I will perform ablation studies by removing identity-related words from training data to evaluate their impact on model bias. I will also perform dialect-specific evaluation by comparing model performance on text written in dialectal forms versus standard English. Model comparison studies will examine how traditional machine learning models perform in comparison to transformer-based models with respect to fairness outcomes. Threshold sensitivity analyses will explore how decision thresholds affect subgroup disparities, while data imbalance analyses will evaluate the role of dataset composition in bias patterns. Finally, counterfactual testing will be conducted by replacing identity terms in sentences, such as changing "He is Black" to "He is white," to observe changes in model predictions. These analyses aim to isolate sources of bias and better understand the relationship between dataset composition, model architecture, and ethical outcomes.

7 Expected Ethical Contribution

This project contributes to ongoing discussions in ethical AI by examining how automated moderation systems may unintentionally reinforce systemic bias. Rather than focusing solely on maximizing predictive accuracy, this work emphasizes fairness and the societal implications of deploying NLP systems in real-world settings. The findings may provide insight into whether hate speech classifiers disproportionately silence marginalized communities, how dataset design influences fairness outcomes, and what technical strategies may help reduce bias in automated content moderation systems.

References

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